

JUNE EXAM AND SUPPLEMENTARY SCOPE, etc / JUNIE EKSAMENEN HEREKSAMEN OMVANG, ens.

THE SCOPE:

Chapters 1, 2, 3 and 4 – Implicitly

Chapter 5 to Chapter 9, including examples and theory development in the textbook

Chapter 10, 11, 15, 16, 18, 19, 20 to Chapter 9, including examples and theory development in the textbook

Everything that we have done in class

Tutorials and tutorial tests are included

EXCLUSIONS:

6.4 Just read the development of the two models of motion in the presence of resistive forces; **BUT**, study model 2: Resistive force proportional to speed squared.

6.3 Motion in accelerated frames

9.8 Deformable systems

9.9 Rocket propulsion

10.9 Rolling motion of a rigid body (Section 10.9)

Everything after example 11.1 is excluded

15.5 Torsional Pendulum (chapter 15) and Sections 15.6 Damped and 15.7 Forced oscillations

16.5 Rate of energy transfer by sinusoidal waves in strings.

16.6 The linear wave equation

18.4 Resonance and the rest of the chapter

20.5 The first law of thermodynamics and the rest of the chapter

REQUIRED DERIVATIONS.

1. Derive the equations of motion (all of them) for constant acceleration using the definition of average acceleration and average velocity using both algebraic and calculus methods.

2. Derive equations 4.8 and 4.8 on page 82, the constant acceleration equations of motion in 2-D

3. Section 4.3 Projectile motion

(Horizontal range and maximum height derivations, equations 4.13 and 4.12)

4. Section 4.6 Relative velocity and relative acceleration

Show that the acceleration of the particle measured by an observer in one frame of reference is equal to that measured by any observer in a reference frame moving with constant velocity relative to the first frame.

5. Work-Kinetic Energy Theorem

Sect 7.5, Work-Kinetic Energy Theorem, Eqn 7.14 to 7.17. You can also use the alternative (no integration) formulation done in both English and Afrikaans classes to arrive at the same result.

6. Systems / Situations involving Kinetic Friction Sect 8.3 page 222 to eqn 8.15

7. Showing that momentum conservation for two particles follows from Newton II and III p. 248

8. Inelastic and Elastic collisions 1D in Ch 9, page 257 – 259 eqn 9.14 – 9.24. (You need not know how to derive eqn 9.21 and 9.22, but note how it is done and note these equations too and be able to determine what happens in extreme cases from eqn 9.21 and 9.22).
9. Collisions in 2 D, page 264 eqn 9.25 – 9.27
10. Derive equation 10.24 in Section 10.7 Rotational Kinetic Energy
11. Motion of an object attached to a spring (section 15.1)
 Analysis Model: Particle in simple harmonic motion (section 15.2)
 Equations 15.1 – 15.18
12. Simple and Rigid pendulums
13. Superposition and standing waves, chapter 18
 Superposition of Sinusoidal waves, page 535 – 536
14. Standing waves, Section 18.2
 Derive Equations 18.1 – 18.3 (page 539)
- 15 Work and heat in thermodynamic processes, section 20.4
 Derive Equation 20.9

THEORY:

Highlighting certain of the theory sections here does not mean that the other sections are not important! It does not exclude them from the test. There may be a bit of repetition between items in “Theory” and in “Required Derivations”

1.
 Chapter 5, Examples 5.9 The Atwood machine and Example 5.10, Acceleration of two objects connected by a cord
 Section 5.8 Forces of friction
 Experimental determination of μ_s and μ_k
 Section 6.1 Extending the particle in uniform circular motion model
 Conical Pendulum
 Banked roadways
 Riding the Ferris wheel
 Section 6.2 Nonuniform circular motion
 Section 6.4 Motion in the presence of resistive forces
 Model 1 (just read through)
 Model 2 (in details as was done in class)
2. Sect 7.5, Work-Kinetic Energy Theorem, Eqn 7.14 to 7.17. You can also use the alternative (no integration) formulation done in both English and Afrikaans classes to arrive at the same result. (Explaining how energy conservation follows from W-KE theorem.
- Sect 7.6, Potential energy of a system, Equations 7.18 to 7.20
- Relationship between a conservative force and the potential energy function, Eqn 7.29 (Explaining how energy conservation follows from W-KE theorem and definition of potential energy. Know definition of potential Energy.)
3. Systems / Situations involving Kinetic Friction Sect 8.3 page 222 to eqn 8.13 & 8.15

4. Changes in mechanical energy for non-conservative forces. Eqn 8.16, page 227

5. Showing that momentum conservation for two particles follows from Newton II and III p. 248

Inelastic and Elastic collisions 1D in Ch 9, page 257 – 259 eqn 9.14 – 9.24. (You need not know how to derive eqn 9.21 and 9.22, but note how it is done, note these equations too and be able to determine what happens in extreme cases from eqn 9.21 and 9.22).

6. Collisions in 2 D, page 264 Equations 9.25 – 9.27

7. Table 10.1 Kinematic equations for rotational and translational motion (provide for yourself in your cheat sheet); and Table 10.2 Moments of inertia of homogenous rigid objects with different geometries (a table will be provided to you)

8. Centre of mass, section 9.6 & Systems of particles, section 9.7. Study the accompanying examples. Discrete cases only: algebraic summation only, no integrals.

9. Calculating moments of inertia, section 10.6. Discrete cases only: algebraic summation only, no integrals.

But, you should be able to use the parallel axis theorem! Equation 10.22

10. Rotational Kinetic Energy, section 10.7

11. Energy Considerations in Rotational Motion

(Look at the equivalence with translational motion. Look at Table 10.3)

12. Section 11.1 The vector product and torque

13. Section 11.2 (Just read through, but know the definition of $\mathbf{L}$)

(Look at how equation 11.10 was derived)

Example 11.1 Angular momentum of a particle in uniform circular motion

14. Limit chapter 13 to the tutorial questions (or similar questions) that were part of tutorials for chapter 7

15. Motion of an object attached to a spring (section 15.1)

16. Analysis Model: Particle in simple harmonic motion (section 15.2)

Equations 15.1 – 15.18

17. Energy of the simple harmonic oscillator (Section 15.3)

18. Sections 15.4 and 15.5

19. Propagation of a disturbance, Section 16.1

20. Analysis Model: Travelling wave, Section 16.2

Equations 16.3 – 16.12

Note Equations 16.18

21. Reflection and Transmission, Section 16.4

22. Waves in chapter 16

23. Superposition and standing waves, chapter 18

- 24. Waves under boundary conditions
 - Study examples 18.3 and 18.4
- 25. Chapter 19
 - Temperature and the Zeroth law of thermodynamics, section 19.1
 - The Celsius, Fahrenheit and absolute temperature scales, page 572, up to and including Example 19.1
- 26. Thermal Expansion of Solids
 - Equation 19.4 (length up to volume)
- 27. Macroscopic description of an Ideal Gas, Section 19.5
 - Equations 19.7 – 19.11
- 28. Chapter 20
 - Heat and Internal Energy, section 20.1 (including units of heat and the mechanical equivalent of heat)
 - Specific heat and calorimetry, section 20.2 (equations 20.3 -20.5)
 - Latent Heat, section 20.3 (including equation 20.6 - 7)
 - Work and heat in thermodynamic processes, section 20.4
 - Equation 20.9